

StratoONE Chat & Files Workspace Module

End-to-End System Integration, Security & Status Code Audit Report

Date: September 12, 2026 | Target Lead: Tech Lead (TL) | Author: Saifan

1. Executive Summary & Working Structure

This report documents the completion, architectural validation, and security testing of the StratoONE Chat and Files Workspace Module. Inspired by Slack's productivity design, the module delivers real-time bidirectional communication, instant optimistic messaging, global Ctrl+K search, a dedicated Files Explorer linked to PostgreSQL/R2, and a comprehensive user status & sign-out workflow.

2. Protocol Status Code Verification (100% Passed)

HTTP Status	Endpoint / Operation	Audit Verification & Details	Result
200 OK	GET /	Next.js SSR application renders workspace layout	PASS
200 OK	GET /api/trpc/room.list	Authenticated user rooms queried with member metadata	PASS
200 OK	GET /api/trpc/upload.listFiles	Live PostgreSQL query returns real files including Saif Ali docs	PASS
200 OK	GET /api/socketio	Socket.IO engine ready for WebSocket & polling transport	PASS
200 OK	GET /...Report.pdf	Static documents & assets served reliably	PASS
403 FORBIDDEN	GET /api/trpc/message.list	Non-member blocked from snooping on unauthorized room	PASS
403 FORBIDDEN	POST /upload.initiate	Blocked unauthorized R2 presigned upload URL creation	PASS
404 NOT FOUND	GET /invalid-route	Route boundary correctly responds with standard 404	PASS
101 SWITCHING	WS Handshake (Socket.IO)	HTTP upgraded to real-time duplex WebSocket stream	PASS

3. Frontend to Backend Working Structure

- Zero-Lag Optimistic Messaging:** When a message is sent, ChatWindow immediately appends it to the React local state with a temporary ID. Concurrently, WebSocket emits message:send. Upon database transaction commit in PostgreSQL, the ack swaps the temporary ID without causing UI flicker or input lock.
- Live Files Explorer Integration:** The Files Explorer is wired to the backend tRPC upload.listFiles endpoint. All items shown (e.g. Md_Saif_Ali_Performance_Review.docx, Saif_Ali_Product_Architecture_v2.pdf) are authentic database records with presigned download links, star toggling, and ownership filters.
- Workspace Top Bar & Global Search (Ctrl+K):** A unified search bar searches channels, messages, and team members with arrow navigation and instant jump-to actions.
- User Profile & Account Popover:** Bottom-left interactive popover showing avatar, display name, handle (@saif.ali), custom status presets, Active/Away toggle, notification pause, and Sign Out.

4. Security Architecture (Authentication & Authorization)

- Authentication (AuthN):** Centralized in src/server/services/auth.ts. Headers (Authorization / x-user-id) are validated on every tRPC batch call and verified by io.use() during WebSocket connection handshakes. Ready for plug-and-play enterprise SSO (Okta, Auth0, Clerk, Azure AD).
- Authorization (AuthZ):** Guarded by protectedProcedure middleware. Room membership is strictly verified in PostgreSQL before reading messages or generating Cloudflare R2 presigned upload/download URLs.
- Storage Protection:** No direct S3 credentials exposed to the client. All files use signed, time-limited tokens generated only for authorized participants.

VERIFICATION SUMMARY: ALL AUDIT CRITERIA SATISFIED

The module has been thoroughly verified across all architectural layers. Zero regressions detected. Status codes (200, 403, 404, 101) behave strictly according to specification. The codebase is production-ready and submitted for final review by the Technical Lead.